

Smart Messenger Feature Documentation



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Overview

Smart Messenger is a powerful, enterprise-ready messaging solution built natively on the Salesforce Platform. It seamlessly integrates **WhatsApp** and **SMS** communication into Salesforce, enabling businesses to engage with their customers in real time — right from Leads, Contacts, Accounts, or any standard/custom object.

Designed with productivity, compliance, and user experience in mind, Smart Messenger combines modern messaging tools with Salesforce data, AI-powered features, and automation to drive faster and smarter conversations

Key Highlights

- **Two-Way Messaging:** Send and receive messages directly within Salesforce — supports both WhatsApp and SMS channels.
- **Real-Time Notifications:** Get live alerts via platform events through the utility bar for every inbound message.
- **Einstein AI Assistance:** Draft better responses, generate summaries, and get smart suggestions using built-in AI tools.
- **Media Support:** Share images, PDFs, videos, and more with rich preview support.
- **Admin Configuration:** Connect Meta (WhatsApp Business) or Twilio (SMS) using a secure, permission-based configuration screen.
- **Template Creation:** Build and manage WhatsApp templates manually or generate them instantly with AI.
- **Workflow Integration:** Embed the chat experience into your lead-to-case lifecycle, marketing flows, or service automation.
- **Responsiveness:** Mobile-friendly UI with support for Salesforce Console Apps, Utility Bar, and Experience Cloud pages.

1. Configuration

Overview

The Configuration module allows Salesforce administrators to securely set up messaging service provider details for channels such as **SMS (Twilio)** and **WhatsApp (Meta/Facebook)**. This forms the foundation for enabling outbound and inbound messaging functionality in the app.

Key Capabilities

- Configure and manage connection settings for:
 - **Twilio (SMS)**
 - **Meta WhatsApp Business API**
- Separate configuration blocks for each channel.
- Role-based access: Only users with the SmartMsg_Admin permission set can modify settings.
- Encrypted handling of sensitive fields (e.g., Auth Tokens, App Secret).
- Channel-based dynamic field rendering (e.g., Twilio fields shown for SMS, Meta fields for WhatsApp).
- Toggle active/inactive status for each provider.
- Real-time validation and user feedback via toast notifications.

Configurable Fields

Channel	Field Name	Description
SMS (Twilio)	Account SID	Twilio account identifier
	Auth Token	Twilio API key
	Messaging Service SID	Twilio messaging service ID
	Twilio Phone Number	Sender number
	Media Base URL	Base URL for media hosting
	Status Callback URL	URL to receive message status updates
	Environment ID	Twilio project environment
	Service ID	Twilio Service ID
WhatsApp (Meta)	Application ID	Facebook App ID
	App Secret Key	Facebook App secret
	Auth Token	Access token
	Phone Number ID	WhatsApp sender number ID
	WhatsApp Business ID	Linked WhatsApp Business Account
	Active	Enable/Disable the channel

Access Control

- Only users assigned the **SmartMsg_Admin** permission set can access or modify the settings.
- Unauthorized users will see an error message indicating lack of permission.

User Experience

- Settings are managed via a Lightning Web Component with a modal-based form for each provider.
- Records are fetched and cached using Apex with `@AuraEnabled(cacheable=true)` annotation.
- Real-time field-level validation with visual feedback ensures accurate configuration.
- Configurations are stored in **Custom Settings** (`SmartMsg__Meta_Configuration__c` and `SmartMsg__TwilioConfiguration__c`) ensuring org-wide access and security.

2. Chat Application(Smart Messenger)

The Chat Application allows Salesforce users to send and receive real-time messages via **WhatsApp** and **SMS**, directly from any standard or custom record like Leads, Contacts, or Accounts. It is designed to be intuitive, responsive, and powerful, supporting multimedia messages, templates, AI capabilities, and more.

Permission-Controlled Access

Only users with the **SmartMsg_Admin** permission set can access the chat component. If a user lacks this permission, they will be shown a friendly message indicating insufficient access. This ensures that sensitive communication functionality is available only to authorized personnel.

Select Communication Channel (SMS / WhatsApp)

Users can choose the desired messaging channel by clicking the icon-based dropdown:

- **SMS Icon** or **WhatsApp Icon** is shown at the footer.
- Once selected, all subsequent messages are sent through the chosen channel.
- WhatsApp messages are enhanced with rich capabilities like media and templates.

Message Composition & Sending

- The user types a message in the input box at the bottom.
- A **phone field selector dropdown** allows choosing the recipient number field from the record (e.g., MobilePhone, Phone).
- A **Send** icon triggers message delivery.
- Character counter is displayed for SMS (500 characters max) and WhatsApp message has no character limits.

Message Scheduling

Users can schedule a message for future delivery:

- Click the **clock icon** to open the schedule panel.
- Choose **date** and **time**.
- Once confirmed, the message is saved with a “Scheduled” tag.
- Scheduled messages are styled differently in the chat with a **clock icon** and time indicator.
- Supports both **SMS** and **WhatsApp** scheduling via backend Apex classes.

Attach Media or Documents

Users can attach a file or a public URL to any WhatsApp message:

- **Click the Attach icon** to open the modal.
- Toggle between:
 - Upload file: .png, .pdf, .mp4, .mp3, etc.
 - Enter public media URL

- Previews are rendered inline in the chat for media files.
- File chips appear below the message box before sending and can be removed.

Request Location

- For WhatsApp, a **“Request Location”** button can be used.
- Sends a predefined message asking the customer to share their location.
- Once the user shares their location, it shows a **map thumbnail** and **link to view location** on Google Maps.
- Useful in logistics, service calls, or delivery workflows.

Message Templates

- Click the **Template icon** to launch the template picker.
- Search templates by name.
- View **template name**, **message body**, and **channel** in a table layout.
- Click on a template to auto-fill it into the message box.
- Especially useful for WhatsApp's 24-hour session rule.

24-Hour Re-Engagement Handling (WhatsApp)

- If the customer hasn't messaged in the last 24 hours, WhatsApp blocks freeform messages.
- The app detects this and shows a **Re-engagement Popup**.
- It prompts the user to send a **pre-approved WhatsApp template** instead.
- Ensures compliance with WhatsApp's policy and avoids message failures.

Pin / Star Important Messages

- Hover over any message and click the **three dots icon** to:

- **Pin** a message – shown in a banner at the top.
- **Star** a message – marked with a star icon.
- Starred and pinned messages help in highlighting key parts of a conversation.

Search Messages

- Click on **More Options > Search Messages**.
- Enables an input box where users can enter keywords.
- Filters and highlights all matching messages in real-time.

Download Chat Transcript

- Available under **More Options**.
- Downloads the full conversation history in a readable format.
- Useful for compliance, audits, or hand-offs.

Inline Media Rendering

Messages with media are rendered visually:

- **Images:** Preview thumbnail with click-to-zoom.
- **PDF / MP4 / MP3:** Rendered with appropriate file icons.
- **Audio:** Playable via file link.
- **Fallback icons** appear if the file is not previewable.

Einstein AI Assistant

The Chat Application includes a built-in **Einstein AI-powered assistant** that helps users communicate more efficiently and professionally. These features are accessible via a toggleable Einstein icon at the bottom of the chat interface and are designed to **boost agent productivity** and **improve message quality**.

How to Access

- Click the **Einstein icon** near the message input box.
- A pop-up menu will appear with the following three options:
 - **Reframe**
 - **AI Summary**
 - **AI Suggestion**

Reframe Message

- This feature allows the user to **rewrite the drafted message** in a more clear, professional, or polite tone using AI.
- It's helpful when the agent is unsure about how to phrase a response or wants to sound more empathetic or persuasive.
- **Use case:** A sales rep drafting a follow-up can get a better-worded version with a single click.

AI Summary

- Instantly generates a **summary of the full conversation** visible in the chat window.
- Great for understanding context before taking over a conversation or when needing to log notes.
- The summary can be **copied and reused** in notes, emails, or tickets.
- **Use case:** A support supervisor reviewing escalation history can quickly grasp the situation.

AI Suggestion

- Recommends **next best responses** based on the ongoing conversation thread.
- AI analyzes the recent incoming message and suggests short, relevant replies.
- Helps new agents or busy users reply faster with minimal thinking or typing.

- **Use case:** If a customer asks for pricing, the AI may suggest, "Sure, I can share our pricing plans. Would you prefer monthly or yearly billing?"

Typing Indicator (WhatsApp Only)

- Automatically sends a typing signal to the customer via the WhatsApp API when the user starts composing a message.
- Simulates human-like interaction and improves engagement.

Real-Time Updates via Platform Events

- The chat listens to SmartMsg__Incoming_Message__e event.
- New incoming messages from the customer appear **instantly**, without refresh.
- Keeps agents updated in real-time for faster replies.

Delivery & Status Tracking

Each WhatsApp/SMS message includes:

- Sent, Delivered, Read indicators via icons.
- Failed messages are marked with a **warning icon** and a tooltip showing the failure reason (e.g., invalid number, 24-hour window expired).

Mobile Responsive Design

The component is fully mobile-optimized:

- Collapsible menus, icon-based actions, and responsive modals.
- Smart layouts for file upload, search, and conversation history.
- Tailored Einstein AI and template views for mobile screens.

3. Template Creation

The **WhatsApp Template Creation** module allows users to create, preview, and manage **pre-approved WhatsApp message templates** that can be reused across conversations. It supports both **manual creation** and **AI-assisted generation**, making it easy for users to build compliant, well-structured message templates.

SMS Templates

SMS templates are created through a simplified form tailored to SMS channel requirements.

Fields include:

- **Template Name:** Unique name for the SMS template.
- **Message Body:** The core message to be sent; supports merge fields like {1}, {2} for runtime data replacement.
- **Placeholder Count:** Displays how many variables are used.
- **Channel:** SMS.

Manual Template Creation for WhatsApp

Users can manually define all parts of a WhatsApp template using a structured form.

Key Input Fields:

- **Template Name:** A unique identifier for the template. Auto-uppercase formatting applied.
- **Language:** WhatsApp language code (e.g., en, hi) used to set the template's locale.
- **Header Type:**
 - Options include:
 - **TEXT:** For short headers (e.g., "Order Update")
 - **IMAGE, VIDEO, DOCUMENT:** Allows media-rich headers
- **Header Text:** The text value for TEXT type headers.
- **Body:** Main message content. Supports variables in the form of {{1}}, {{2}}, etc.
- **Footer Text:** Optional, used for sign-off or disclaimers.

- **Button Type:**

- **Quick Reply:** Up to two buttons (e.g., "Yes", "No")
- **URL:** Buttons that open external links
- **Phone Number:** Initiates a call

Form Features:

- **Real-time Validation:** Ensures required fields are filled and structured correctly.
- **Auto Cleanup:** Resets fields after submission or cancel.
- **Previews:** Live preview panel shows how the final template will look on WhatsApp.

AI-Powered Template Suggestions

A standout feature of this module is the **AI assistant integration**, which helps users auto-generate template content using Einstein-powered logic.

AI Features:

1. **Generate Template via AI:**

- a. Users provide a **basic intent** or **message description** (e.g., "appointment confirmation").
- b. AI returns a complete template structure with:
 - i. Suggested header
 - ii. Formatted body with variables (`{{1}}`, `{{2}}`)
 - iii. Optional footer
 - iv. Button recommendations
- c. Fields auto-populate based on the AI output.
- d. Enhances productivity and helps non-technical users craft WhatsApp-compliant templates.

2. **Auto-fill Buttons:**

- a. Based on message context, AI may suggest:
 - i. Predefined replies
 - ii. CTA buttons (e.g., "View Order", "Call Support")
 - iii. URL suggestions using dummy or placeholder links

3. **Multi-Language Support (Planned):**

- a. The AI generation is adaptable for multiple WhatsApp languages in future releases.

Template Editing & Management

- **Edit Existing Templates:**
 - Templates can be retrieved from Salesforce and edited (WhatsApp Templates can be edited only once in a 24hours).
 - Upon selection, the form is auto filled with saved values.
- **Save & Cancel:**
 - Save persists the template into the custom object (SmartMsg__WhatsApp_Template__c).
 - Cancel resets the form and hides the modal.

4. Incoming Message Window (Utility Bar)

The **Incoming Message Window** is a powerful real-time notification panel integrated into the Salesforce Utility Bar. It allows users to view, manage, and respond to inbound WhatsApp and SMS messages instantly without navigating away from their current workspace. This tool boosts agent productivity and ensures no customer message goes unnoticed.

Real-Time Platform Events Listener

- The component listens to SmartMsg__Incoming_Message__e Platform Event.
- As soon as a new inbound message is received, it:
 - Plays a notification sound using Web Audio API.
 - Blinks the Utility Bar icon to draw attention.
 - Automatically fetches and displays the message without refresh.

Record Navigation

- Clicking on the sender's name or message header navigates to the associated Contact, Lead, or Account record using **NavigationMixin**.
- It uses the matched record from SOSL lookup based on the incoming phone number.

Tabs: All & Unread

- Two-tab filters:
 - **ALL**: Displays all inbound messages grouped by sender, sorted by recent activity.
 - **UNREAD**: Displays only unread message threads for easy triage.

Search Messages

- A search bar is available to filter messages by:
 - Sender name
 - Message body content
- Live filtering works across tabs with pagination preserved.

Message Preview & Media Rendering

- Messages grouped by sender and show the latest message content.
- Rich preview support for media:
 - Images: Inline thumbnails
 - Audio: Embedded player
 - Videos: Inline video player
 - Files (PDF, DOC, etc.): Download link with file icon

Channel Identification

- Each message displays an icon for its channel:
 - WhatsApp: Custom WhatsApp icon
 - SMS: Salesforce's standard SMS icon

Message Actions (per thread)

Each message thread includes:

- **Reply:** Opens the full Chat Application prefilled with the record and number.
- **Mark as Read:** Clears unread badge, stops blinking, and updates unread count.
- **Delete:** Permanently deletes the message with confirmation. Supports server-side deletion via Apex.

Pagination Support

- Built-in controls for Previous and Next page navigation.
- Pagination is calculated based on the filtered and grouped message list.

Refresh Capability

- A refresh button reloads all messages.
- Animated icon gives feedback while loading new messages.

Utility Bar Integration

- Utility Item Label updates dynamically:
Example: **Incoming Messages (3)**
- Highlighted (blinking) utility icon indicates a new message.

- Stopped blinking after:
 - Marking as Read
 - User interaction (Reply or Delete)

Quick Access to Full Chat

- Clicking **Reply** or a sender opens the **Chat Component** directly inside the utility view.
- Chat loads with full functionality, including AI tools, templates, and media support.
- A **Back button** lets users return to the message list.

Last Sent User Awareness

- Uses ***getLastSentUser*** Apex method to check if the current user sent the last outbound message to the sender.
- Utility Bar highlight stays active if the same user is expected to respond, ensuring accountability.

Mark All as Read (Per Sender)

- When a thread is marked as read, all messages from that sender are updated via the Apex method **markAsRead**.

Permission Controlled Backend

- All data operations are wrapped with **WITH SECURITY_ENFORCED** to ensure FLS and CRUD compliance.
- Apex classes handle errors gracefully and show toasts on UI for feedback.

